Dashboard Design & Interactive Streamlit Specification

Global Olympic Sports Performance Analytics Dashboard (Snowflake + Streamlit)

Date: 24 September 2026	**Team ID:** PNT2026TMID092418
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 5 Marks

Activity 1: Interactive and Visually Appealing Dashboards

Creating an effective Streamlit / Tableau dashboard involves presenting historical Olympic performance data stored in Snowflake through clear, interactive, and intuitive visual elements. Key architecture guidelines applied include:

• **Clear and Intuitive Layout:** Structured top-level metric KPI cards (Total Medals, Total Athletes, Top Yield Sport, Lead Region Efficiency) followed by interactive charts organized in logical grid columns.

• **Use Appropriate Visualizations:** Horizontal bar charts for per-athlete yield ranking, bubble charts for investment vs. medal efficiency, multi-line charts for 1992–2026 medal trends, and donut charts for regional distribution.

• **Colour and Theming:** Professional dark blue and slate palette (`#1A365D`, `#2B6CB0`) with high-contrast accent colors for medals (Gold, Silver, Bronze) and regional indicators.

• **Interactive Filters and Slicers:** Sidebar drop-down selectors for Olympic Year (1992–2026), Season (Summer/Winter), Geographic Region, Sport Category, and Country Delegation.

• **Drill-Down Capabilities:** Click-through functionality enabling users to select a sport category (e.g., Rowing) and drill down into individual discipline event results and per-athlete metrics.

• **Responsive Design:** Fluid grid layout engineered to adapt seamlessly across desktop workstations, tablet devices, and web browser windows.

• **Custom Visuals and Icons:** Visual indicators for host nation status, medal icons, and efficiency percentage badges.

• **Use of Infographics:** Executive summary banner highlighting overall Olympic performance trends and investment efficiency ratios.

Major Analytical Outcomes & Key Dashboard Insights

Here are 5 core analytical outcomes derived from the Snowflake-powered Streamlit and Tableau Olympic analytics dashboard:

1. Rowing and Shooting emerge as highest per-athlete medal yield categories: Rowing leads with **6.916 average medals per athlete**, followed closely by Shooting at **6.731**. This proves that sports with multi-event disciplines or high technical density deliver the highest medal ROI per qualified athlete for National Olympic Committees (NOCs).

2. Global medal performance is led by top athletic powerhouses: The **United States leads overall cumulative totals with 735 medals**, followed by **China with 696 medals**. These benchmarks reflect deep sports infrastructure, comprehensive athlete development programs, and sustained long-term capital investment.

3. Capital efficiency is highest in South American athletic programs: The Regional Efficiency chart demonstrates that **South America leads global efficiency at 49.64%**, followed by Europe/Asia at **45.98%**. Overperforming nations such as Slovakia, Georgia, and Ethiopia deliver disproportionately high medal counts relative to lower baseline sports budgets.

4. Host nation advantage provides a measurable performance boost: The Host Dominance bar chart shows **Japan leading home host medals with 105**, followed by the United States with **100 host medals** and China with **93 host medals**, confirming a consistent 25–40% performance bump when competing on home soil.

5. Olympic medal totals and gender parity expanded from 1992 to 2026: The multi-decade trend line illustrates total medals awarded fluctuating from a low of **305 in Sydney 2000** to a peak of **363 in Tokyo 2020**, while 2026 recorded **334 medals** alongside near 50-50 gender participation parity across events.

Overall Dashboard Summary: The interactive Snowflake dashboard proves that Olympic success is driven by strategic capital allocation in high-yield sports (Rowing/Shooting), targeting capital to high-efficiency regions (South America), leveraging home nation advantage, and adapting to modern multi-decade participation expansion.